Newtonian Spacetime Is Curved!

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1 Motivation

This is a physics lecture, making use of the auto-parallel equation of Lecture 8. The claim: in the standard formulation Newtonian spacetime is not curved, but if Newton's axioms are inspected properly then Newtonian spacetime must be considered *curved*, and this curvature absorbs the effect of the gravitational force, so that gravity can no longer be regarded as a force. This has all to do with gravity and nothing to do with relativity: it is not relativity that forces us to spacetime, but the understanding of the relation between geometry and gravity.

2 Newton's axioms, read carefully

Newton wrote his laws in words (and drawings). Without the usual addition "in an inertial system" (which has to do with choosing coordinates), they read:

N1. A body on which no force acts moves uniformly along a straight line.

N2. The deviation of a body's motion from such uniform straight motion (today: the acceleration) is effected by a force, reduced by the reciprocal of the body's inertial mass: the larger the inertial mass, the less the same force accelerates the body.

N1 is not a special case of N2. If N1 is read as a statement about what a particle does, it is the special case $F = 0$ of N2, and Newton would not have needed to state it separately. For N1 to play a role of its own it must be read differently:

Remark. N1 must be read as a **measurement prescription for the geometry of space**: for particles that are known to be subject to no force, whatever path they follow *defines* what a straight line (uniform straight motion) is. (Later we will add something to "space".) This is sensible: a straight line is obvious on paper, but in nature it has to be defined operationally.

N2 then tells us what happens to particles on which a force acts, for instance because they carry charges and react to fields (the modern Faraday point of view). In the language of Lecture 8: a straight line is an auto-parallel, $\nabla_v v = 0$, and the *deviation* from it, the acceleration, is $\nabla_v v \neq 0$, which is again a vector. If one does not know what the straight lines are (the connection coefficients Γ), one cannot even calculate the acceleration.

The problem with gravity. Gravity acts on *every* massive particle. In a universe with at least two particles, the second particle exerts a gravitational force on the first at arbitrarily large distance, so the assumption of N1 (no force acts) can never be met. Hence:

If N1 is to remain applicable, gravity must not be considered a force.

We do not know how to encode gravity other than as a force, so there is a serious problem. (One could discard N1, but then "deviation from a straight line" would have to refer to the straight lines of Euclidean geometry, and one would have to explain why physical space has Euclidean geometry. So we rather try to save N1; this is a fully conservative approach, nothing in it contradicts Newton's *Principia*.)

3 Laplace's question

Laplace (1749–1827) asked:

> Can gravity be encoded in a curvature of space, such that its effects show if particles under the influence of no other force are required to move along straight lines in this curved space?

A science-museum demonstration (a coin rolling in a funnel-shaped surface) suggests this: the curved two-dimensional surface and the coin's orbits look like planetary orbits one dimension higher. The answer is a resounding **no**, and the demonstration is entirely wrong; but it is instructive to see why, because the idea can be repaired.

3.1 The standard view: gravity as a force

Write the Newtonian equation of motion for a particle in the gravitational field

$$m\,\ddot{x}^\alpha = F^\alpha(x), \qquad \alpha = 1, 2, 3.$$

The gravitational force field is determined by the mass density ρ of matter via the Poisson equation, with $F^\alpha = -\partial^\alpha \phi$ for the potential ϕ:

$$\nabla^2 \phi = 4\pi G \rho,$$

which is the field equation of Newtonian gravity. A characteristic fact is that the gravitational force is proportional to the mass of the body *on which it acts*. Then m cancels and the motion depends on the body not at all:

$$\ddot{x}^\alpha = f^\alpha(x), \qquad f^\alpha := F^\alpha/m \text{ (gravitational force per mass)}.$$

Theory does not say this is true; experiment decides (this is called the **weak equivalence principle**): in a tube with a feather and a bead, the bead falls faster in air, but in an evacuated tube both fall identically, within the measurement error. All of this can be attributed to air resistance.

3.2 Why Laplace's idea fails in space

Interpret the equation as an auto-parallel equation: bring f to the other side and ask whether

$$\ddot{x}^\alpha - f^\alpha(x) = 0$$

can have the form $\ddot{x}^\alpha + \Gamma^\alpha{}_{\beta\gamma}(x)\,\dot{x}^\beta \dot{x}^\gamma = 0$ for some connection coefficients Γ on three-dimensional space. It cannot: the Γ depend only on position, but the terms they multiply are quadratic in the velocity, while $f^\alpha(x)$ is independent of velocity. No choice of Γ can match them (for any velocity, including double the velocity). So Newton's equation for a gravitational force cannot be written as an auto-parallel equation: there is no notion of straight line in space that absorbs the effect usually attributed to a force. Laplace found other inconsistencies instead, which come down to the same obstruction.

A remark on pointwise removal. One can make the acceleration vanish at one point by going to the coordinate system that falls with the body (the Γ can be made to vanish at a point, Lecture 7). But a homogeneous gravitational field can be transformed away entirely, whereas an *inhomogeneous* one cannot: on the round Earth, a frame falling at the North Pole does not remove the effect for a freely falling body at the South Pole.

4 Repairing Laplace's idea: Newtonian spacetime

Laplace did not read N1 fully: it speaks of *uniform* motion along a straight line, not just motion along a straight line. A curve is more than the set of its points: it is the set of its points together with its *parameterization*. Straight but non-uniform motion is not straight uniform motion. We introduce the appropriate data structure to keep track of the difference, one that is already familiar from school (the position-versus-time diagram of a harmonic oscillator): attach a time axis. This introduces no new data, merely represents the existing data in another way (no relativity involved).

In spacetime, one does not need to say "uniform and straight motion in space": straight uniform motion in space is simply a straight curve in spacetime, because the parameterization is stored in the extra coordinate. A non-uniform motion (equal time intervals covering decreasing distances) is drawn as a curve that is visibly not straight in the spacetime diagram.

Example 1 (World lines)**.** A trajectory $x : \mathbb{R} \to \mathbb{R}^3$, $t \mapsto x(t)$, is converted back and forth into a *world line*

$$X : \mathbb{R} \to \mathbb{R}^4, \qquad t \mapsto \big(t,\, x^1(t),\, x^2(t),\, x^3(t)\big),$$

the history of the particle. A bound planetary orbit becomes a spiral in spacetime; projecting out the time component recovers the orbit. Nothing new has been introduced.

4.1 First attempt, in coordinates

Assume x satisfies $\ddot{x}^\alpha = f^\alpha(x)$ and let $X = (x^0, x^1, x^2, x^3)$ be the world line with $x^0(t) = t$ (indices a, b, c run $0, \ldots, 3$; Greek indices α, β, γ run $1, 2, 3$). Then

$$\dot{x}^0 = 1, \qquad \ddot{x}^0 = 0.$$

Hence the equations

$$\ddot{x}^0 = 0, \qquad \ddot{x}^\alpha - f^\alpha(x) = 0$$

(the first trivially true, and doing no harm). Multiplying the second equation's f term by $(\dot{x}^0)^2 = 1$ turns them into the single set of equations

$$\ddot{x}^a + \Gamma^a{}_{bc}\, \dot{x}^b \dot{x}^c = 0, \qquad a = 0, \ldots, 3,$$

provided

$$\boxed{\Gamma^\alpha{}_{00} = -f^\alpha(x), \qquad \text{all other } \Gamma^a{}_{bc} = 0}$$

(a term with $\dot{x}^1 \dot{x}^1$, say, in the α equation would not have the required form; $\Gamma^0{}_{bc}$ must vanish so that $\ddot{x}^0 = 0$). So *in this chart*, Newton's equation is an auto-parallel equation in four-dimensional Newtonian spacetime, i.e. Laplace's idea works there. (Only $\Gamma^\alpha{}_{00}$ is non-zero, and it is determined by the gravitational force per mass.)

Not a coordinate artifact. One must check that this is not just a feature of the chart: could the Γ be made to vanish in another chart? That depends on the curvature. With the Γ above, the only non-vanishing components of the Riemann tensor are

$$R^\alpha{}_{0\beta 0} = -\,\partial_\beta f^\alpha,$$

the *tidal force tensor* (minus the Hessian of the gravitational potential, up to sign conventions); this follows from $R^i{}_{jmn} = \partial_m \Gamma^i{}_{jn} - \partial_n \Gamma^i{}_{jm} + \Gamma^i{}_{sm}\Gamma^s{}_{jn} - \Gamma^i{}_{sn}\Gamma^s{}_{jm}$. Contracting gives the 00 component of the Ricci tensor,

$$R_{00} = -\partial_\alpha f^\alpha = \nabla^2 \phi = 4\pi G \rho,$$

by the Poisson equation. Hence:

- The Poisson equation is the statement that the (Ricci) curvature of Newtonian spacetime is sourced by the mass density: $R_{00} = 4\pi G\rho$ (for dust, this is the 00 component of the relativistic form of Einstein's equations, to which this is the non-relativistic limit; the idea, due to Einstein around 1912, that spacetime Ricci curvature is the energy-momentum of matter, appears here already, though the relativistic equations contain an extra, trace term that is missing in this naive form).
- Where $\rho \neq 0$ the curvature does not vanish, and by the theorem of Lecture 7/8 there is then no chart in which all Γ vanish. So the effect is not a coordinate artifact.
- The curvature is a curvature of *spacetime*, not of space: $R^{\alpha}{}_{\beta\gamma\delta} = 0$ for all spatial indices. The only non-vanishing components have time indices. Space remains flat; spacetime is curved in the extra direction together with space.

4.2 Conclusion and geometric statements

Laplace's idea works on Newtonian spacetime. The first axiom now reads: a particle under the influence of no force other than gravity (which no longer counts as a force) has a world line that is a straight line, an auto-parallel, *in spacetime*. Converging of initially parallel auto-parallels (two particles dropped side by side towards the Earth) is another sign of curvature, called *geodesic deviation* (as parallel lines from the equator on a sphere eventually meet). The space is flat, but the spacetime is curved.

What is the acceleration? In the chart, the world line equation has the shape $\ddot{x}^a + \Gamma^a{}_{bc}\dot{x}^b\dot{x}^c$. The first term alone is not the components of a vector, and neither is the second (the Γ do not transform as a tensor, and neither does $\ddot{x}^a$); only the *sum* cancels the troublesome terms and transforms as a vector. This sum

$$a^a = \ddot{x}^a + \Gamma^a{}_{bc}\dot{x}^b\dot{x}^c$$

is the components of the acceleration vector $\nabla_{\dot{X}}\dot{X}$. One cannot call $\ddot{x}^a$ "the acceleration" and the rest an "extra acceleration": the acceleration is chart independent (sitting in a rocket, you feel whether you accelerate or not, whatever coordinates someone else thinks of).

5 The geometric formulation

Everything above was in charts. The foundations of the geometric formulation (not new physics: what Newton wrote down, implemented properly) are as follows. The lecture only sketches them.

Definition 1 (Newtonian spacetime)**.** A *Newtonian spacetime* is a quintuple $(M, \mathcal{O}, \mathcal{A}, \nabla, t)$ where $(M, \mathcal{O}, \mathcal{A})$ is a four-dimensional smooth manifold, ∇ a connection and $t : M \to \mathbb{R}$ a smooth function (the *absolute time*), satisfying

(i) $\mathrm{d}t \neq 0$ at every point of M (this gives absolute space, see below);

(ii) $\nabla \mathrm{d}t = 0$ (the covariant derivative of the gradient of absolute time vanishes: absolute time "flows uniformly");

(iii) ∇ is torsion-free.

(The lecturer first forgot (iii) and added it later: one condition on the time function, one on the connection, and a compatibility condition between the two.)

Spacetime is not "space plus time". M is just a four-dimensional manifold in which at first one cannot distinguish space from time (like the two directions on a sphere). One must delete all memory of what spacetime "is". Space and time are derived notions.

Absolute space. Define *absolute space at time* τ as the level set $S_\tau = \{p \in M \mid t(p) = \tau\}$. Since $\mathrm{d}t \neq 0$ everywhere, M is the disjoint union of the S_τ: spacetime *foliates* into spaces, with the universal clock reading the same on each one. If $\mathrm{d}t$ vanished somewhere, level surfaces could meet at that point, "time would stop" there, which is excluded.

Uniform flow of time. $\nabla \mathrm{d}t = 0$ is written with the first slot of ∇ unfilled; it is a $(0,2)$ tensor field required to be zero. In a chart this says $(\nabla_a \mathrm{d}t)_b = \partial_a (\mathrm{d}t)_b - \Gamma^c{}_{ba} (\mathrm{d}t)_c = 0$, which constrains the connection coefficients.

Definition 2 (Time orientation of vectors)**.** A vector $X \in T_pM$ is

- *future directed* if $\mathrm{d}t(X) > 0$,
- *spatial* if $\mathrm{d}t(X) = 0$ (tangent to the level surface of t),
- *past directed* if $\mathrm{d}t(X) < 0$.

Newton's laws, geometrically.

- **N1:** the world line of a particle under no force (gravity not counting as one) is a *future directed auto-parallel* of ∇: $\nabla_v v = 0$ and $\mathrm{d}t(v) > 0$ everywhere on the curve (a particle runs into the future, not the past).
- **N2:** the acceleration of a world line is a force:

$$\nabla_v v = \frac{F}{m},$$

  where F is a *spatial vector field* ($\mathrm{d}t(F) = 0$), since a force is only supposed to accelerate in spatial directions. F may of course vary from point to point in spacetime (for instance, a rocket engine switched on and off is captured, because the field differs at different points). It just has no component pointing in the time direction. Nobody said $m\,a = F$ is wrong; the statement is that the acceleration is $\nabla_v v$, not $\ddot{x}$.

5.1 Stratified atlases and recovering chart physics

As a convention (very practical, but nothing depends on it), restrict to *stratified atlases*: charts (U, x) whose zeroth coordinate function coincides with the absolute time function, $x^0 = t|_U$. This is always possible since t is globally defined. In such a chart, $(\mathrm{d}t)_b = \delta^0{}_b$ is constant, so $\nabla \mathrm{d}t = 0$ becomes

$$\Gamma^0{}_{ab} = 0 \quad \text{for all } a, b.$$

(In a non-stratified atlas one would get extra terms, which are coordinate artifacts.)

Take a world line X with arbitrary parameter λ (primes denote $\mathrm{d}/\mathrm{d}\lambda$) satisfying N2 and split the equation into components:

- $a = 0$: $x^{0\prime\prime} = 0$ (since $\Gamma^0{}_{bc} = 0$ and $F^0 = 0$). Hence $x^0(\lambda) = a\lambda + b$ with constants: the absolute time along a world line runs proportionally to the parameter. One may therefore parameterize world lines by absolute time, $\lambda = t$, and then primes become dots (up to a constant factor a^2 on the left, absorbed into the choice of time units on the force side).
- $a = \alpha$: splitting the sum over the lower indices into spatial and temporal ones, and using torsion-freeness ($\Gamma^\alpha{}_{\gamma 0} = \Gamma^\alpha{}_{0\gamma}$, which produces a factor 2),

$$\ddot{x}^\alpha + \Gamma^\alpha{}_{\beta\gamma} \dot{x}^\beta \dot{x}^\gamma + 2\,\Gamma^\alpha{}_{\gamma 0}\, \dot{x}^\gamma + \Gamma^\alpha{}_{00} = \frac{F^\alpha}{m}.$$

This is Newton's second law written out. Its terms have the usual interpretation, but only the whole left-hand side is *the* acceleration; none of its terms is an acceleration by itself:

- $\Gamma^\alpha{}_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma$ involves only spatial indices: it is the correction when a curvilinear spatial chart (e.g. polar coordinates) is used while time is left untouched;
- $\Gamma^\alpha{}_{00}$ is velocity independent: in a rotating chart it is the centrifugal pseudo-acceleration, and in the presence of gravity it contains the gravitational force $-f^\alpha$;
- $2\Gamma^\alpha{}_{\gamma 0}\dot{x}^\gamma$ is linear in the spatial velocity: the Coriolis pseudo-acceleration. The factor 2 comes from the absence of torsion.

Example 2 (Inertial versus rotating charts)**.** In the absence of gravity, Newtonian spacetime is flat: there is a chart (an inertial frame) where all Γ vanish, and N2 reads $\ddot{x}^\alpha = F^\alpha/m$, as usual. In a chart that rotates in space (a single fixed chart on spacetime), the Γ are non-zero and the individual terms above do not vanish; but their sum is unchanged, because the acceleration of a particle is a vector and does not care about our fantasy of a rotating coordinate system. Pseudo-accelerations are called "pseudo" because separately they are not accelerations.

Remark on the first part versus this part. The first half of the lecture started from preconceptions and charts to reach an insight; this part is a fresh start with the geometric definition, from which the chart statements are *derived*. In particular no flatness of ∇ is assumed anywhere; only $\mathrm{d}t \neq 0$, $\nabla \mathrm{d}t = 0$ and torsion freeness. (On the question whether torsion affects auto-parallels: this is notoriously confusing and depends on how the connection is built from given structures; here torsion-freeness is simply postulated, as in relativity, where the connection is also required to be metric compatible.)

Remarks from questions.

- A spatial metric on each absolute space is an inertial-system fiction: in non-inertial systems the geometry is richer than a 3-metric (a rotation co-vector Ω enters too). The Lagrangian with kinetic energy $\frac{1}{2}m\,\dot{x}^2$ (no square root) needs such a spatial metric and works only over spatial sections; this does not carry over to spacetime.
- In general relativity the Lorentzian metric philosophically replaces the time function (light cones instead of level surfaces), not the spatial metric.

Summary

- N1 only plays a role if read as a measurement prescription for straight lines; but since gravity acts on every mass, gravity cannot count as a force if N1 is to stay applicable.
- Laplace's attempt (a curved *space* whose straight lines absorb gravity) fails: $\ddot{x}^\alpha = f^\alpha(x)$ is not of auto-parallel form.
- Adding the time coordinate repairs it: with $\Gamma^\alpha{}_{00} = -f^\alpha$ and all other $\Gamma = 0$, Newton's equation becomes an auto-parallel equation of *spacetime*. The Riemann tensor has only $R^\alpha{}_{0\beta 0} = -\partial_\beta f^\alpha$ (tidal tensor), so it is not a chart artifact, space stays flat, and the Poisson equation becomes $R_{00} = 4\pi G\rho$.
- Geometric formulation: a Newtonian spacetime is $(M, \mathcal{O}, \mathcal{A}, \nabla, t)$ with $\mathrm{d}t \neq 0$, $\nabla \mathrm{d}t = 0$, torsion-free ∇. N1: force-free world lines are future-directed auto-parallels; N2: $\nabla_v v = F/m$ with F spatial. In stratified charts this reproduces the usual equations, with the pseudo-accelerations as separate, non-vector Γ terms whose sum is the chart independent acceleration.